INCH-POUND

MIL-PRF-25369D
w/AMENDMENT 1
30 March 2007

SUPERSEDING
MIL-PRF-25369D
1 May 1998

PERFORMANCE SPECIFICATION

CARTRIDGE, INFLATOR, FLOTATION GEAR, (CARBON DIOXIDE), (FOR THE INFLATION OF PNEUMATIC LIFE PRESERVERS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.

This specification covers puncturable-seal, carbon dioxide (CO_2) filled cartridge cylinders for the inflation of pneumatic life preservers.

1.2 Classification.

The carbon dioxide cylinders shall be of the following types, as specified:

Type I - 25 to 28 gram capacity.

Type II - 28 to 31 gram capacity.

Type III - 34 to 37 gram capacity.

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2. APPLICABLE DOCUMENTS

2.1 General.

The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Non-Government publications.

The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ-Z 1.4 Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available from www.asq.org or the American Society for Quality, 600 North Plankinton Ave., Milwaukee WI 53203-2914 or www.ansi.org or the American National Standards Institute, 25 West 43rd St., 4th Floor, New York NY 10036.)

2.3 Order of precedence.

In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article.

When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials.

All materials shall be of the corrosion-resistant type, suitably treated to resist corrosion due to salt spray or any other atmospheric condition likely to be met in storage or normal service. See 3.2.1 if any type of plating or coating is to be used to protect against corrosion.

3.2.1 Material limits for coated or plated threads.

When externally threaded parts are to be coated or plated, the minimum pitch diameter may be reduced by an amount equal to 0.3 times the pitch diameter tolerance but not more than 0.0015 inch. All external thread elements shall be within the adjusted tolerance before coating. The limits for coated or plated threads shall be the same as required for uncoated or unplated threads.

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3.3 Recycled, recovered, or environmentally preferable materials.

Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle cost.

3.4 Gas charge.

Filled cylinders shall contain non-medicinal carbon dioxide that shall conform to table I, when tested as specified (see 4.7.1):

TABLE I. Properties of non-medicinal carbon dioxide.

PROPERTY	REQUIREMENT	CO_2 CAPACITY
Purity, Minimum percent by volume	99.5%	Type I: ≥25 and ≤28 grams
Acidity	0-6.8	Type II: ≥28 and ≤31 grams
Moisture content, maximum	0.092 mg/liter	Type III: ≥34 and ≤37 grams
Odor	No objectionable odor	

3.5 Interface.

Figures 1 and 2 represent general concepts for three types of carbon dioxide cylinders. They should be used for dimensional purposes such as length, width, and thickness, but not construed as the only possible design approach. Type I cylinders shall conform to the general dimensions specified on figure 1. Types II and III cylinders shall conform to the general dimensions specified on figure 2.

3.5.1 Cylinder threads.

The threads shall conform to figure 1 for type I and figure 2 for types II and III cylinders. The acquisition documents shall identify the required threading and verification procedures thereof (see 6.2 and 6.6).

3.5.2 Puncturable area.

Each cylinder shall be provided with a puncturable area as shown on figures 1 and 2.

3.6 Performance.

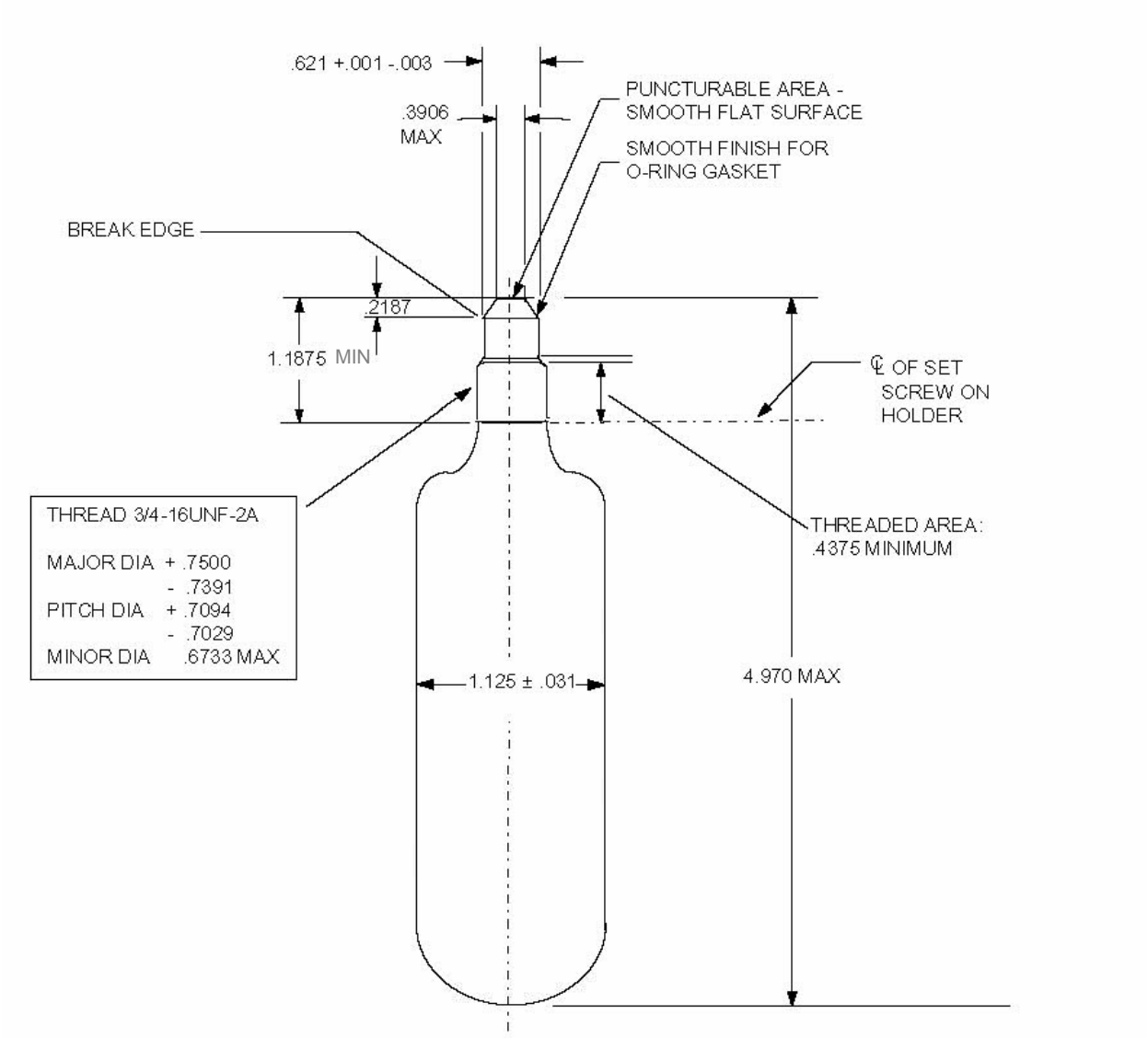
3.6.1 Leakage.

Filled cylinders shall show no evidence of leakage at any point when subjected to the leakage and cyclic leakage tests specified in 4.7.3 and 4.7.7.

3.6.2 Deformation.

Cylinders shall be examined to ensure that there is no deformation of the cylinder or of the puncturable area after being tested as specified in 4.7.5.

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NOTE:

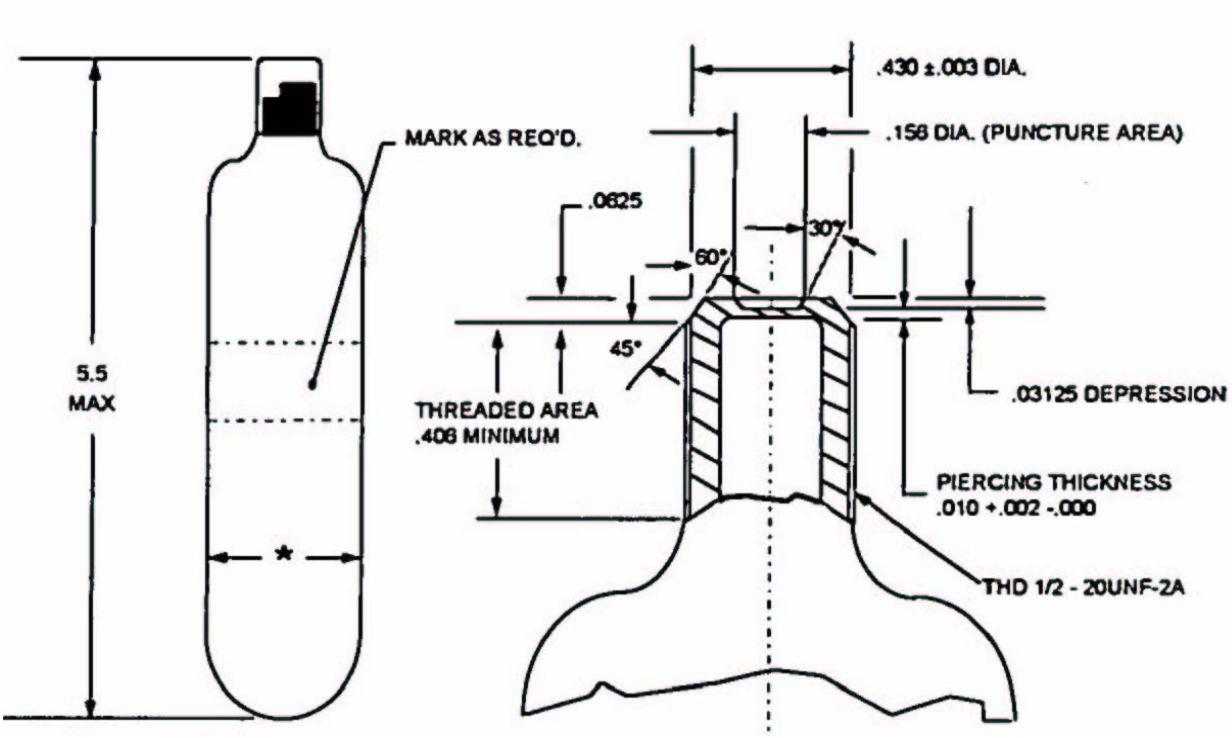
Dimensions are in inches, unless otherwise specified.

Tolerance on:	Fractions	Decimals
	±1/64	±.0156

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FIGURE 1. Carbon dioxide cylinder, type I.

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NOTE:
Dimensions are in inches, unless otherwise specified.

***Diameter of cylinder:**
Type II – 1 inch
Type III – 1-1/8 inches

Tolerance on:	Fractions	Decimals	Angles
	±1/64	±.005	±2º

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FIGURE 2. Carbon dioxide cylinders, types II and III.

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3.6.3 Hydrostatic pressure.

Cylinders shall not burst when subjected to an internal hydrostatic pressure of 8,000 pounds per square inch (psi), when tested as specified in 4.7.8.

3.6.4 Rupture.

When the filled cylinder is subjected to the rupture test specified in 4.7.9, there shall be no splintering or fragmentation upon rupture of the cylinder. Complete separation of the puncturable area or of any portion of the cylinder shall be considered fragmentation.

3.6.5 Maximum filling density.

The maximum filling density of the cylinder shall be 72 percent for type I and 76 percent for types II and III. The filling density is the ratio of weight of carbon dioxide in a container to the weight of water that the container will hold (see 4.7.10 for equation) expressed as a percent.

3.6.6 Minimum gross weight.

Indicated minimum gross weight shall be the empty weight of the cylinder (rounded off to the next lower 1/2 gram), plus the weight of the minimum gas charge.

3.7 Finish.

Cylinders, including threads, shall be treated for corrosion. The exposed surface shall be reasonably smooth, without any sign of bulging or deformation, and shall be free from pit marks and cracks.

3.8 Identification.

The carbon dioxide cylinder shall be marked for identification with the following specific information:

CARTRIDGE, INFLATOR, FLOTATION GEAR (CARBON DIOXIDE)
MIL-PRF-25369
Type (enter I, II, or III)
Minimum Gross Weight: __*__ Grams
Federal Supply Code for Manufacturers (FSCM) and/or Name of Manufacturer ___*___
Date of Manufacture (DoM) (Month/Year): ____*____

*(Information to be entered by the manufacturer)

The identification marking on the cylinder shall be as permanent as the normal life expectancy of the cylinder and capable of withstanding the environmental tests and cleaning procedures specified. The marking method shall be rubber stamp stencil applied over a protective finish before oiling. Use type I, black, No. 37038, ink in accordance with A-A-208. Cover with a clear lacquer. Markings shall be legible and readily visible during normal operational use. Identification marking on identification plates shall be of a color which is in contrast to the color of the surface of the plate. Identification tag marking, when used, shall be permanent to the extent required for the utilization of the item. The use of marking materials that create a hazard are prohibited. The marking of items shall be accomplished in a manner which shall not

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adversely affect the life and utility of the item. Therefore, impression stamping or any other deformation method, when applied directly to the cylinder, is not allowed.

3.9 Workmanship.

The cylinders shall be constructed in accordance with commonly accepted industrial workmanship standards.

4. VERIFICATION

4.1 Classification of inspections.

The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

c. Carbon dioxide inspection (see 4.4).

4.2 First article inspection.

First article inspection shall consist of all tests described under 4.7. If the first article samples are tested by the contractor, 12 cylinders of each type shall be subjected to the first article inspection.

4.3 Conformance inspection.

Conformance inspection shall consist of:

a. Individual tests (see 4.3.1).

b. Sampling tests (see 4.3.2).

4.3.1 Individual tests.

Each cylinder shall be subjected to the following tests:

a. Gas charge (see 4.7.1).

b. Threading inspection (see 4.7.2).

c. Leakage (see 4.7.3).

4.3.2 Sampling tests.

The sampling plans and associated tests specified below are destructive. Test samples submitted by the manufacturer shall not be considered part of the contract or order. Unless otherwise specified, sampling for inspection shall be in accordance with ANSI/ASQ Z1.4. For inspection purposes, a lot shall consist of all cylinders of one type presented for acceptance at the same time. Rejection plans shall be described in the acquisition documents (see 6.2).

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4.3.2.1 Sampling plan A.

Cylinders shall be selected at random from each lot and subjected to the tests listed below and described in 4.7. A minimum of six (6) cylinders shall be selected.

a. Examination of product (see 4.7.4).
b. Deformation (see 4.7.5).
c. Dimensional check (see 4.7.6).
d. Cyclic leakage (see 4.7.7).

4.3.2.2 Sampling plan B.

Three of the sample cylinders selected in accordance with sampling plan A shall be subjected to the hydrostatic pressure (burst) test (see 4.7.8).

4.3.2.3 Sampling plan C.

Three additional samples of the cylinders from sampling plan A shall be subjected to the rupture test (see 4.7.9).

4.3.2.4 Sampling plan D.

Six cylinders shall be selected at random from each lot of 2000 cylinders (or fraction thereof) and subjected to the maximum filling density test (see 4.7.10).

4.4 Carbon dioxide inspection.

The purity, acidity, and moisture content of the non-medicinal carbon dioxide shall be determined in order to verify compliance with table I (see 4.8, 6.7, and 6.8 for recommended test procedures).

4.5 Test conditions.

4.5.1 Atmospheric conditions.

Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches of mercury and at a temperature of 77 ±18 ºF. Where tests are made with atmospheric pressure or temperature substantially different from the above values, proper allowance shall be made for the change in instrument reading.

4.6 Verification alternatives.

The cylinders shall meet all performance and functional requirements specified in section 3. Verification tests indicated in section 4 shall be conducted only after contract execution. Alternative test methods, techniques, or equipment, including the application of statistical process control, or cost-effective sampling procedures can be proposed and used provided these alternatives offer equal assurance that all product requirements are met. Such acceptable alternative verification approaches shall be inserted into the contract as negotiated replacements. In case of dispute as to whether any alternative provides equal assurance, the verification tests in this specification apply.

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4.7 Methods of inspection.

Table II provides a cross-reference matrix of the requirements tested or verified in the paragraphs below.

TABLE II. Requirements cross-reference matrix.

REQUIREMENT	VERIFICATION
3.1	4.2
3.2	4.7.4
3.4	4.7.1
3.5.1	4.7.2
3.5.2	4.7.5, 4.7.9
3.6.1	4.7.3, 4.7.7
3.6.2	4.7.5
3.6.3	4.7.8
3.6.4	4.7.9
3.6.5	4.7.10
3.7	4.7.4
3.8	4.7.4
3.9	4.7.4

4.7.1 Gas charge.

Each cylinder shall be weighed prior to charging with carbon dioxide. The cylinders shall then be charged, sealed, and heated to a temperature of 160 ±2 ºF and maintained at this temperature for a minimum of 20 minutes. After a period of 24 hours has elapsed, cylinders shall be weighed to determine compliance with 3.4.

4.7.2 Threading inspection.

Threads shall be inspected for compliance with the requirements provided in the acquisition documents. Any method of measuring the characteristics of thread forms is acceptable as long as it can be demonstrated and ensures conformance (see 6.6).

4.7.3 Leakage.

After the gas charge has been determined, filled cylinders shall be immersed in a water solution (1% aerosol OT clear, 25 % pure or equivalent) to a depth so that the uppermost portion of the cylinder is at least one inch below the water level for a minimum of two minutes. A 20-inch (mercury) vacuum shall be maintained above the surface of the solution.

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4.7.4 Examination of product.

Cylinders shall be visually inspected to determine conformance with the requirements of this specification with respect to general construction, finish, markings, materials, and workmanship.

4.7.5 Deformation.

There shall be no deformation of the cylinder or of the cylinder puncturable area after heating to a temperature of 160 ±2 ºF, for not less than 20 minutes.

4.7.6 Dimensional check.

Sample cylinders shall be measured to determine compliance with the dimensional requirements of figures 1 and 2. Dimensional checks shall be made following the leakage tests.

4.7.7 Cyclic leakage.

Filled cylinders shall be subjected to two complete temperature cycles. Each cycle shall consist of maintaining the cylinders at -65 ±2 ºF for two hours, and then at 160 ±2 ºF for two hours. After completion of the temperature cycles, cylinders shall be tested for leakage when specified (see 4.7.3 and 6.2).

4.7.8 Hydrostatic pressure (burst) test.

Each sample cylinder shall be labeled to distinguish it from other samples. After the labeling is complete, sample cylinders shall be subjected to an internal hydrostatic force of not less than 8,000 psi at 70 ±2 ºF for a period of not less than five seconds.

4.7.9 Rupture test.

WARNING: This test is dangerous (see 6.3). The sample cylinders shall be tested for fragmentation by gradually heating the cylinder as suggested in 6.3. This test will verify that the ruptured cylinder does not splinter, fragment, or fail at the cap. Fragmentation is defined as the complete separation of the puncturable area or of any portion of the cylinder.

4.7.10 Maximum filling density test.

Sample charged cylinders shall be weighed, then the carbon dioxide discharged and cylinders weighed again. The difference between the first weighing and the second is the quantity of carbon dioxide inside. The cylinder shall then be filled with water and weighed. The ratio of the weight of water to the weight of carbon dioxide is the density, calculated as follows:

$$\text{Filling density} = \frac{\text{weight of } CO_2}{\text{weight of } H_2O} \text{X } 100$$

The maximum filling density for the type I cylinder shall be 72 percent. The maximum filling density for types II and III shall be 76 percent.

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4.8 Acidity.

To test for acidity, pass one liter of the non-medicinal carbon dioxide from the container through 50 milliliters (mils) of recently boiled distilled water which has been cooled to room temperature. Regulate the flow so as to require 15 minutes for delivery of one liter of the gas. The delivery tube shall have an orifice approximately one millimeter (mm) in diameter and shall extend to within two mm of the bottom of the vessel containing the distilled water. The vessel employed shall give a hydrostatic column of 12 to 14 centimeters, with 50 mils of distilled water. The gas shall be measured by means of a gas meter after its passage through the distilled water. After the passage of the gas, pour the liquid into one of two similar comparator tubes. To tube A add two drops of methyl orange indicator. To tube B containing 50 mils of cooled, recently boiled distilled water, add one ml of 0.01N hydrochloric acid solution and two drops of methyl orange indicator. Viewed downward over a white surface, the liquid in tube A showing a deeper shade of red than in tube B will indicate failure of this test.

5. PACKAGING

5.1 Packaging.

For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2.) When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory)

6.1 Intended use.

The carbon dioxide cylinder covered by this specification is intended to provide a means of storing carbon dioxide to inflate a life preserver.

6.2 Acquisition requirements.

Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Type of cylinder and threading required, and procedures for verifying compliance (see 1.2 and 3.5.1).

c. When a first article is required (see 3.2).

d. Whether the cyclic leakage test will be conducted (see 4.7.7).

e. Rejection criteria for tests specified in section 4, if different from requirements specified in section 3.

f. Packaging requirements (see 5.1).

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6.3 First article reports and test samples for Navy procurement.

First article inspection reports and samples for procurement for the Naval Air Systems Command should be forwarded to the Human Systems Department, Quality Assurance Team, Code 4.6.8.3, NAWCAD, Patuxent River MD 20670.

6.4 Suggested equipment for rupture test.

A perforated, 3/8-inch thick, steel container with a bolted top should be used for holding the carbon dioxide cylinder during heating. A can of Sterno™ brand fuel or similar is suggested as a heating mechanism.

6.5 Marking methods.

The tables in MIL-STD-130, “Marking methods” and “Criteria in selection of marking methods”, provide guidance for identification marking.

6.6 Threading of cylinders.

FED-STD-H28/20 shows different inspection methods for thread conformance. Gages should meet requirements specified in FED-STD-H28/6; new or more effective methods are acceptable.

6.7 Purity.

One of two methods may be used for testing: (a) gas chromatography, or (b) Orsat (volumetric) analyzer or equivalent with a capability of determining the purity with an accuracy of at least 0.1 percent. The carbon dioxide will be drawn from the vapor phase of the container. The carbon dioxide container and its contents will be at a temperature between 70 and 75 ºF. At least 90 percent by weight of the original contents will remain in the container after determining the percentage of carbon dioxide. The percentage of carbon dioxide will be calculated by volume. Purity of the non-medicinal carbon dioxide not as specified in table 1 will indicate failure of this test.

6.8 Moisture content.

The moisture content of non-medicinal carbon dioxide can be determined by one of four methods: (a) the accelerated gravimetric method (see 6.8.1), (b) the dew point (frost) method, (c) the electrolytic hygrometer method, or (d) the electrical conductivity method. The carbon dioxide sample will be drawn from the container as a liquid. Vapor phase samples will be conditioned either as described in 6.8.1 or by vaporization within a sampling cylinder before being admitted to the test apparatus. The conditioning procedure will ensure that all of the water present goes into the vapor phase with the carbon dioxide. If vaporized in a sampling cylinder, the sample quantity should not be more than 10 percent of the normal capacity of the sampling cylinder. The sampling cylinder may be gradually heated to evaporate water condensed during the liquid carbon dioxide vaporization. Non-medicinal carbon dioxide containing more than 0.092 milligrams of water per liter of gas, at 760 mm of mercury and 70 ºF, will indicate failure of this test.

6.8.1 Accelerated gravimetric method.

Determine the increase in weight of phosphorus pentoxide (P_2O_5) when subjected to a flow of a measured volume of carbon dioxide as follows:

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Connect the carbon dioxide container to the test apparatus by means of a length of bent, seamless, stainless-steel tubing, a high-pressure union, a needle valve, and a glass-to-metal joint. Rubber connections are not acceptable; however, a sleeve joint secured with Tygon™ or equivalent is acceptable. Connect a mercury safety valve between the low-pressure outlet of the needle valve and the first absorber. Use steel tubing of a fabrication that will have a capability of first passing through a water bath at room temperature and then into a second bath at a temperature above the critical temperature of liquid carbon dioxide. Use an apparatus consisting of three U-tube absorbers in a series, each four (4) inches high and 1/2 inch in diameter, containing P_2O_5 on glass wool, and connected through a water-saturator to a calibrated wet test meter of 1/20-cubic-foot size equipped with a 1-liter dial. Flush the needle valve connecting the container to the absorber with carbon dioxide so that only gas discharged from the container will pass through the absorber.

Bring the absorbers to constant weight in a stream of dry carbon dioxide before weighing, so that when the absorbers are weighed they will be at all times filled with carbon dioxide. Before each weighing of the absorbers, open the absorbers momentarily to the air, and then close, clean, and expose to radiation from a quartz mercury arc for two (2) minutes to remove static charges. Discharge 100 liters through the apparatus at a rate not to exceed 0.4 liter per minute. Calculate the gas volume corresponding to the gain in weight to 760 mm of mercury and 70 ºF. Compute the moisture content in terms of milligrams per liter.

6.9 Subject term (key word) listing.

accelerated gravimetric method

atmospheric pressure

cyclic leakage

dew point (frost) method

electrolytic hygrometer method

electrical conductivity method

gas charge

gravimetric method

hydrostatic pressure

phosphorus pentoxide (P_2O_5)

6.10 Amendment notations.

The margins of this specification are marked with vertical lines to indicate where modifications from this amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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CONCLUDING MATERIAL

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Reviewer:

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